Alternating Current

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Learning Outcomes

In this chapter, you'll learn…

- How **phasors** make it easy to describe sinusoidally varying quantities.
- How to use **reactance** to describe the voltage across a circuit element that carries an alternating current.
- How to analyze an *L*-*R*-*C* series circuit with sinusoidal emfs of different frequencies.
- What determines the amount of power flowing into or out of an alternating current circuit.
- Why **transformers** are useful, and how they work.

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Introduction

- Waves from a broadcasting station produce an alternating current in the circuits of a radio (like the one in this classic car).



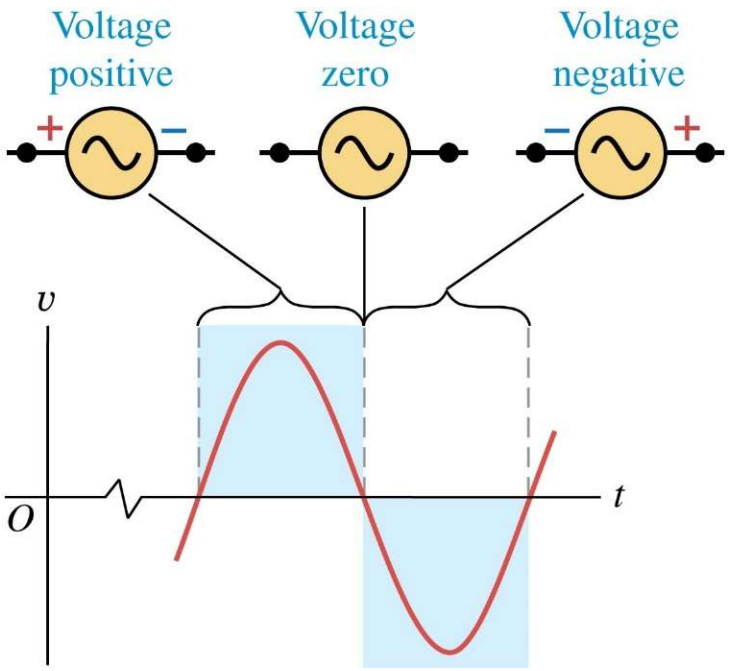
- How does a radio tune to a particular station?
- How are ac circuits different from dc circuits?
- We shall see how resistors, capacitors, and inductors behave with a sinusoidally varying voltage source.

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ac Sources

- Most present-day household and industrial power distribution systems operate with alternating current (ac).
- Any appliance that you plug into a wall outlet uses ac.
- An **ac source** is a device that supplies a sinusoidally varying voltage.



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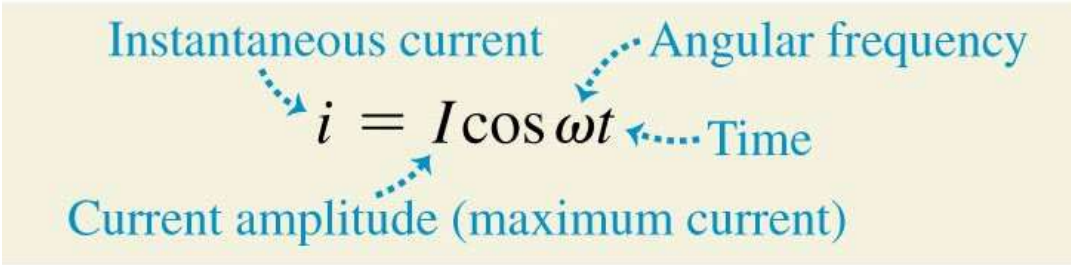
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ac Sources and Currents

- A sinusoidal voltage might be described by a function such as:

$$\upsilon = V\cos\omega t$$

- Here *v* is the instantaneous potential difference, *V* is the voltage amplitude, and $\omega = 2\pi f$ is the angular frequency.
- In the United States and Canada, commercial electric-power distribution systems use a frequency *f* = 60 Hz.
- The corresponding sinusoidal alternating current is:

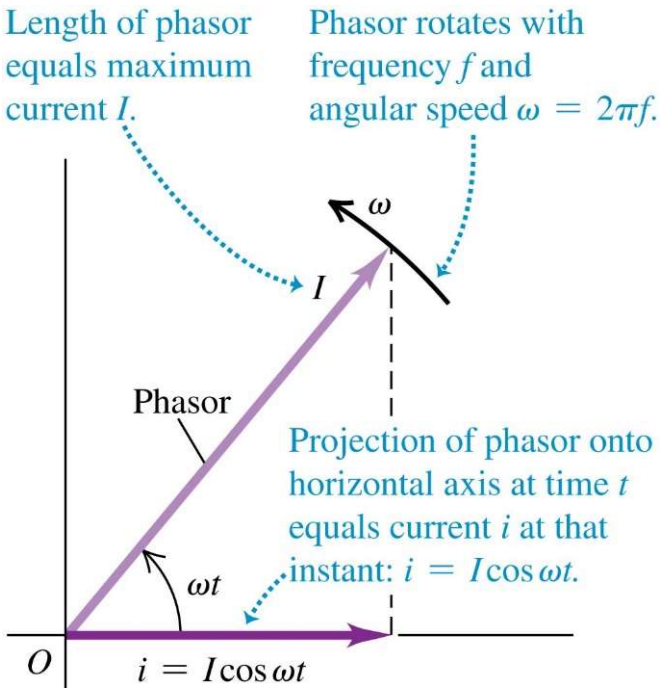


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Phasor Diagrams

- To represent sinusoidally varying voltages and currents, we define rotating vectors called **phasors**.
- Shown is a **phasor diagram** for sinusoidal current.

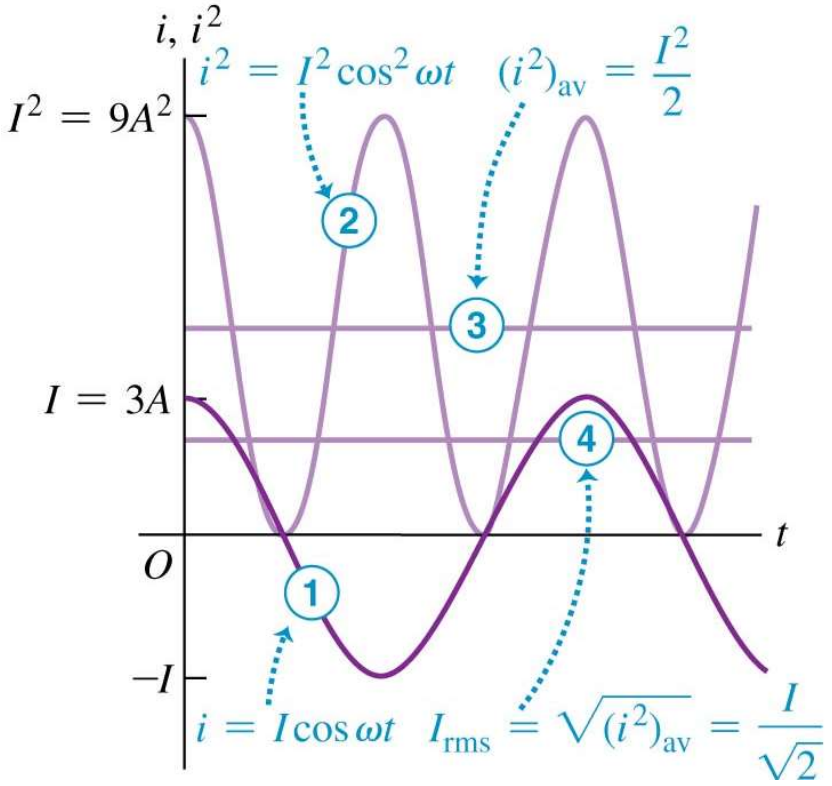


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Root-Mean-Square Values (1 of 2)

- To calculate the rms value of a sinusoidal current:
  1. Graph current *i* versus time.
  2. Square the instantaneous current *i*.
  3. Take the average (mean) value of i^2.
  4. Take the square root of that average.



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Root-Mean-Square Values (2 of 2)

- For sinusoidal ac sources, the rms current and voltage values are:

Root-mean-square (rms) value of a sinusoidal current

$$I_{\text{rms}} = \frac{I}{\sqrt{2}}$$

I: Current amplitude

Root-mean-square (rms) value of a sinusoidal voltage

$$V_{\text{rms}} = \frac{V}{\sqrt{2}}$$

V: Voltage amplitude (maximum value)

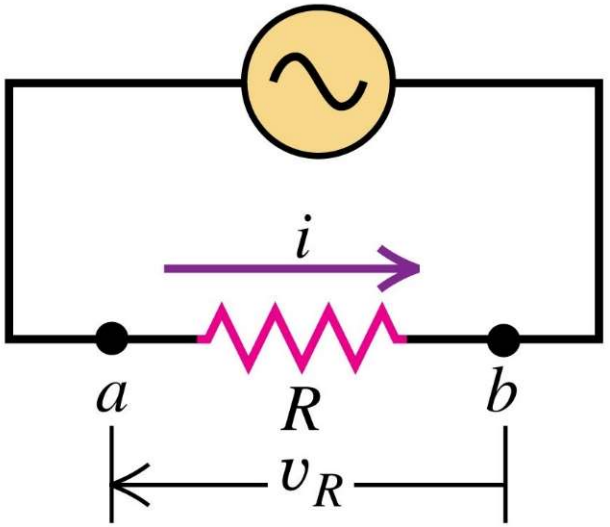
- Usual household power supply in Europe and Australia has a voltage amplitude of V = 325 V, meaning that the voltage alternates between +325 V and −325 V.
- The rms voltage is V_{rms} = 230 V.

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Resistor in an ac Circuit (1 of 3)

- When a resistor is connected with an ac source, the voltage and current amplitudes are related by Ohm's law:



Amplitude of voltage across a resistor, ac circuit

$$V_R = IR$$

I: Current amplitude

R: Resistance

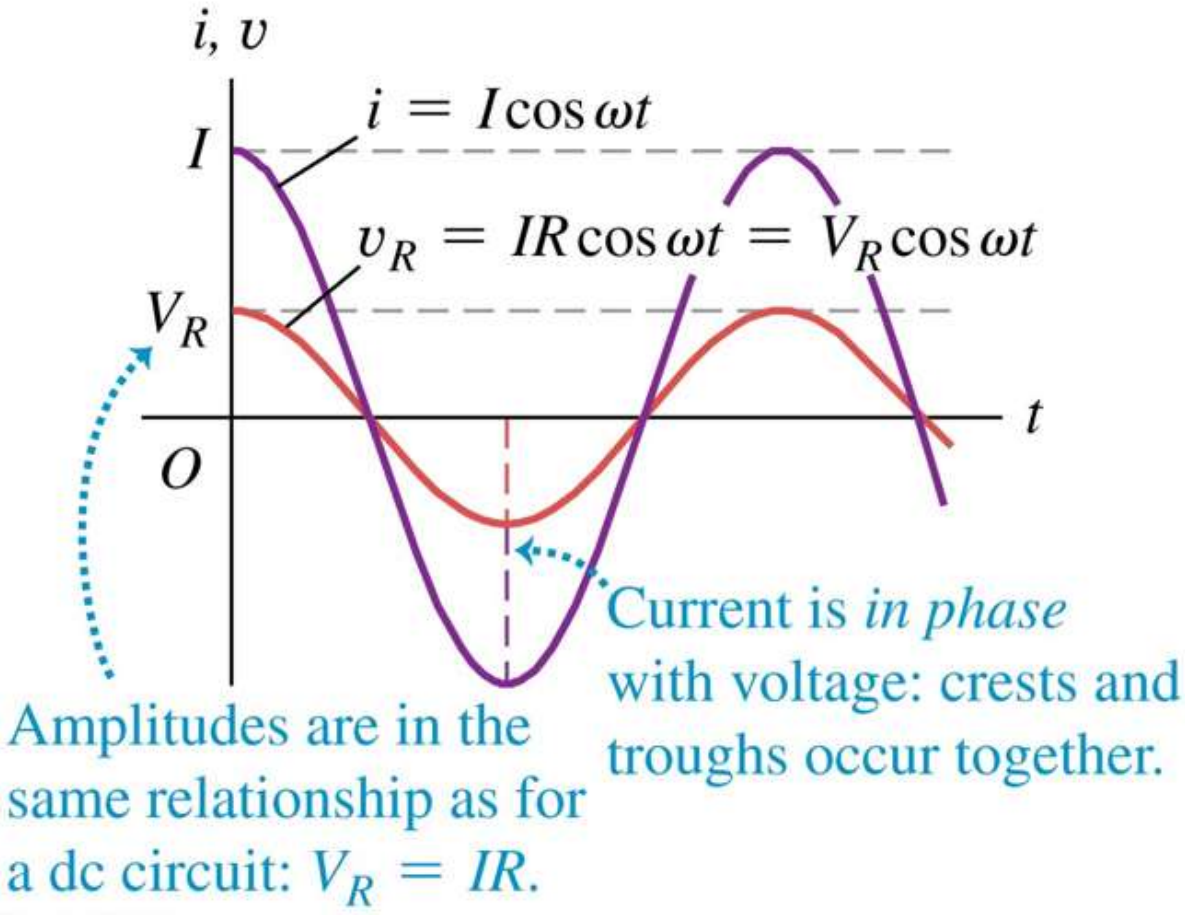
- The resistance does not depend on the frequency of the ac source.

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Resistor in an ac Circuit (2 of 3)

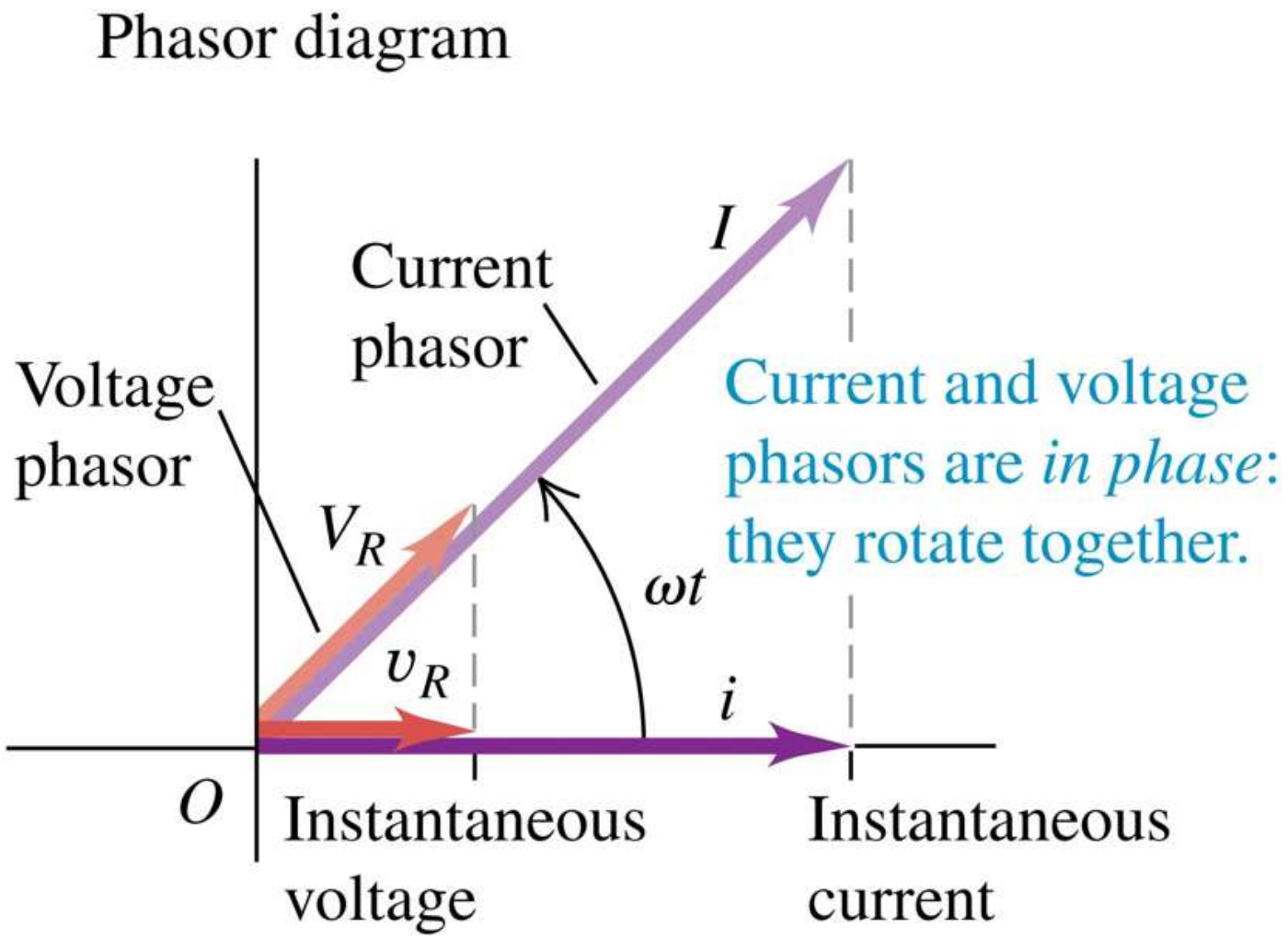
Graphs of current and voltage versus time



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Resistor in an ac Circuit (3 of 3)

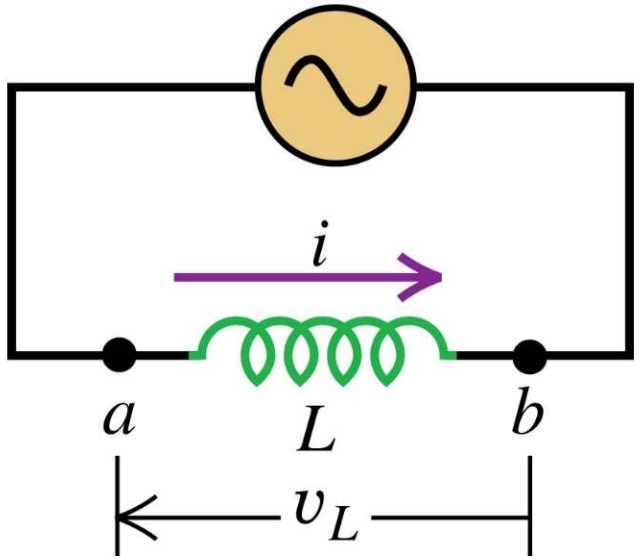


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Inductor in an ac Circuit (1 of 3)

- When an inductor is connected with an ac source, the voltage and current amplitudes are related by:



Amplitude of voltage across an inductor, ac circuit $V_L = IX_L$ (Current amplitude; Inductive reactance)

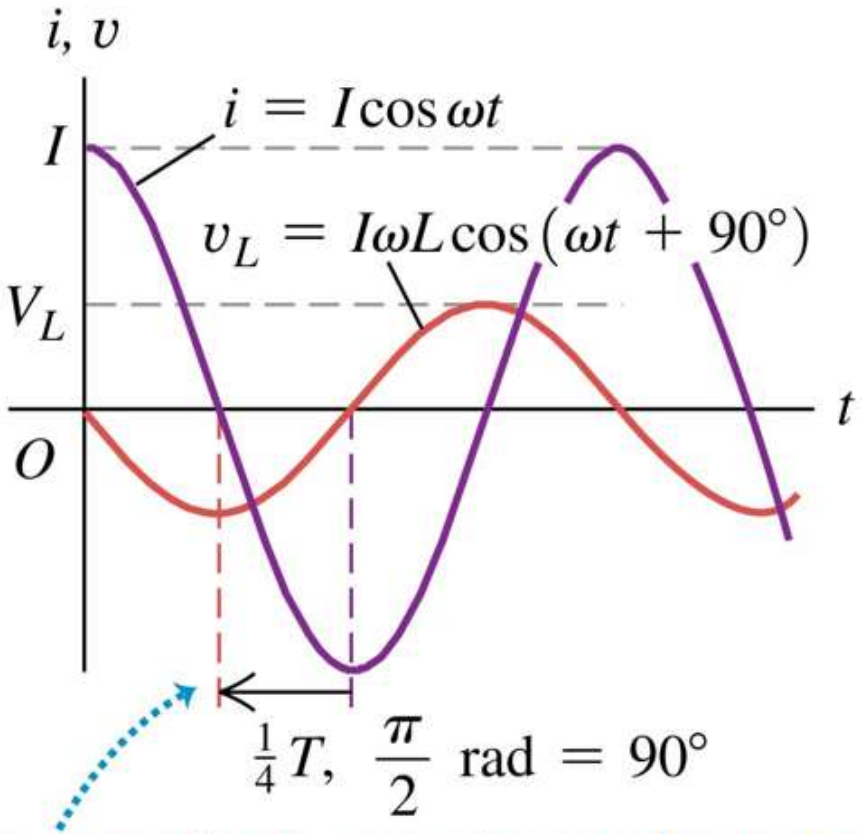
- The inductive reactance is $X_L = \omega L$; the greater the inductance and the higher the frequency, the **greater** the inductive reactance.

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Inductor in an ac Circuit (2 of 3)

Graphs of current and voltage versus time



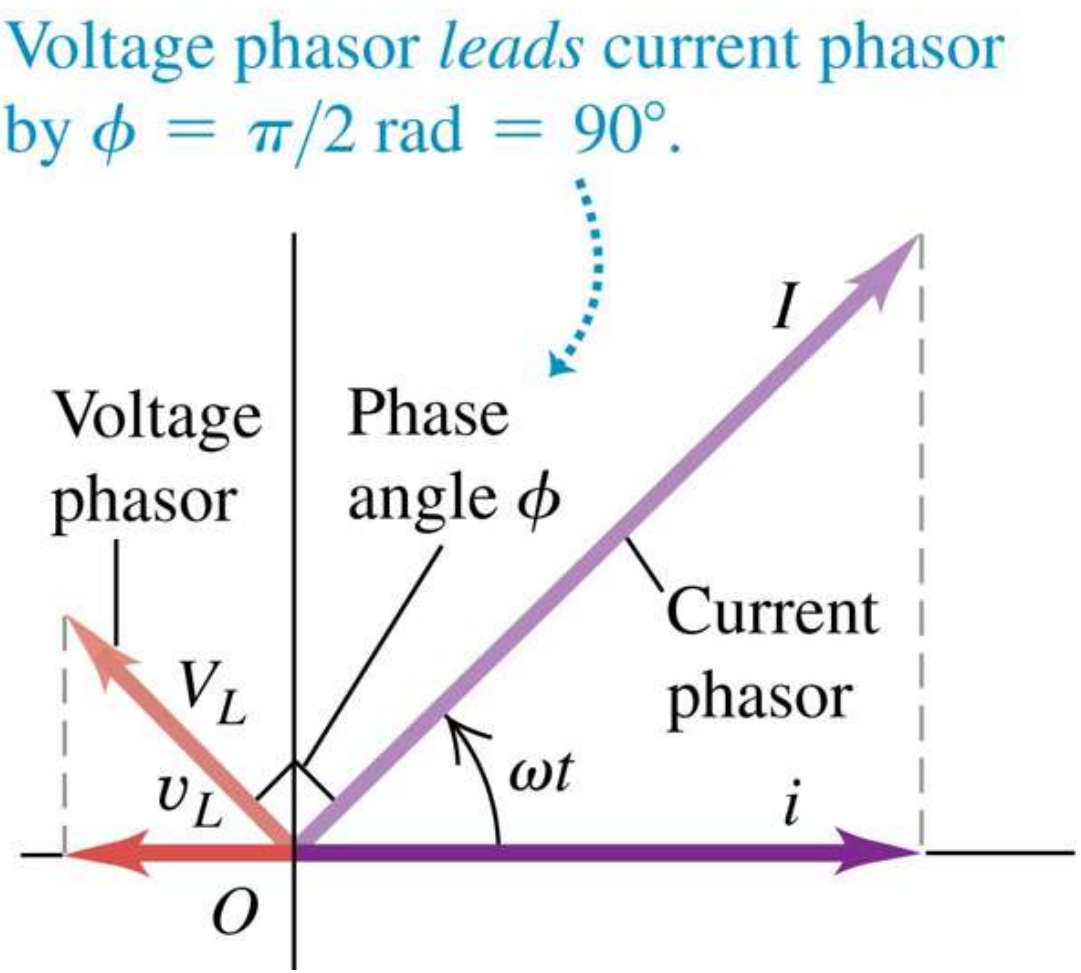
Voltage curve *leads* current curve by a quarter-cycle (corresponding to $\phi = \pi/2$ rad $= 90°$).

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Inductor in an ac Circuit (3 of 3)

Phasor diagram

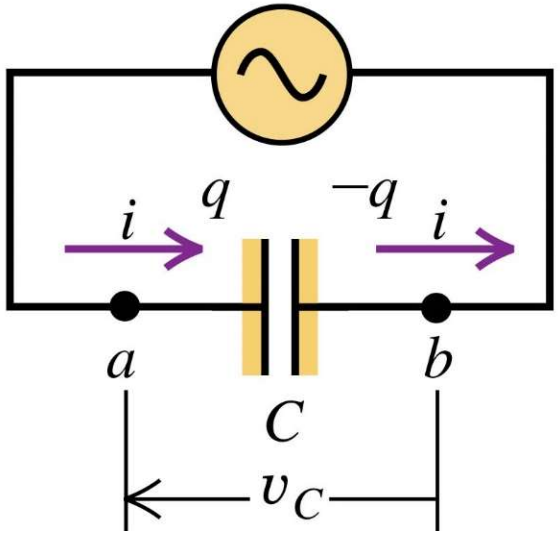


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Capacitor in an ac Circuit (1 of 3)

- When a capacitor is connected with an ac source, the voltage and current amplitudes are related by:



Amplitude of voltage across a capacitor, ac circuit $V_C = IX_C$ (Current amplitude; Capacitive reactance)

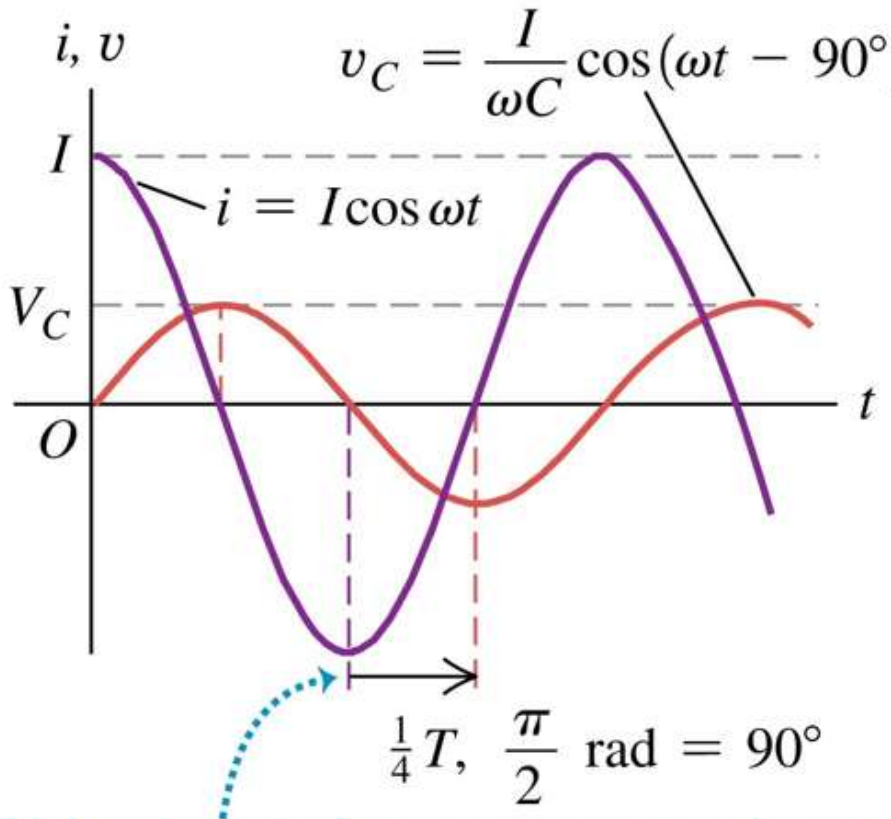
- The capacitive reactance is $X_C = \frac{1}{\omega C}$; the greater the capacitance and the higher the frequency, the **smaller** the capacitive reactance.

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Capacitor in an ac Circuit (2 of 3)

Graphs of current and voltage versus time

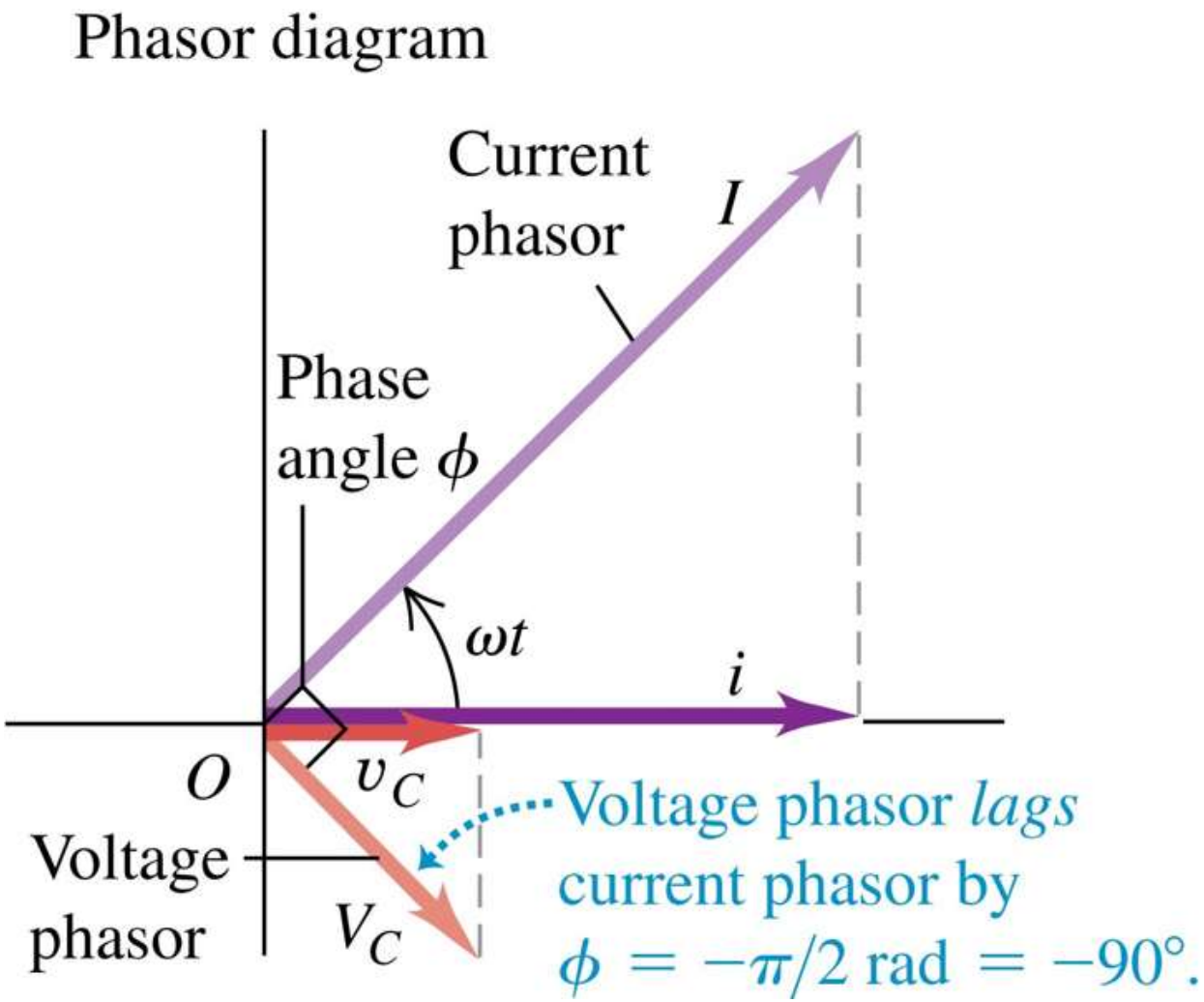


Voltage curve *lags* current curve by a quarter-cycle (corresponding to $\phi = -\pi/2$ rad $= -90°$).

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Capacitor in an ac Circuit (3 of 3)

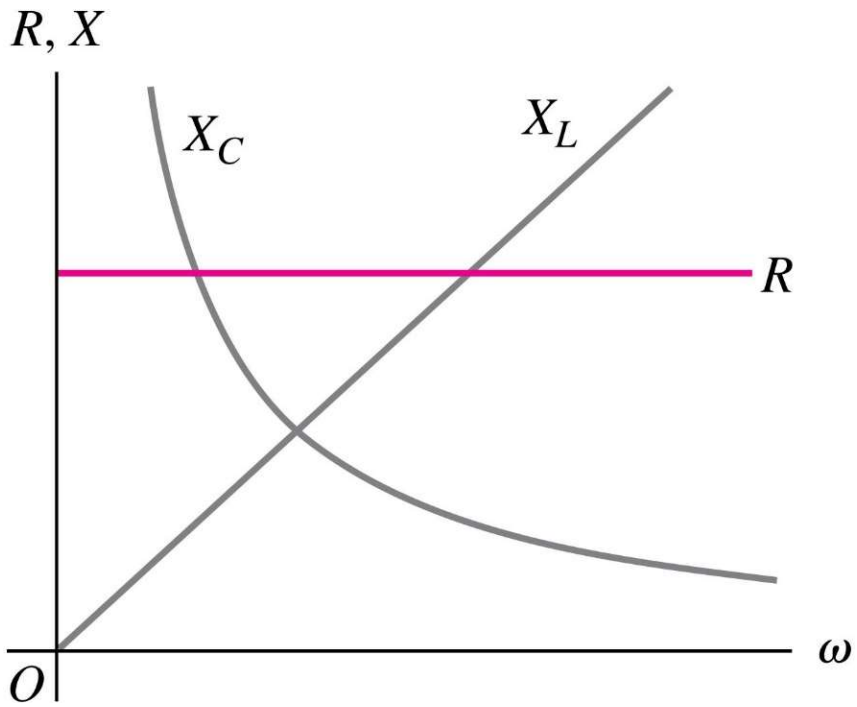


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Comparing ac Circuit Elements

- The graph shows how the resistance of a resistor and the reactances of an inductor and a capacitor vary with angular frequency ω.
- Resistance R is independent of frequency.
- If $\omega = 0$ corresponding to a dc circuit, there is no current through a capacitor because $x_C \to \infty$.
- In the limit $\omega \to \infty$ the current through an inductor becomes vanishingly small.

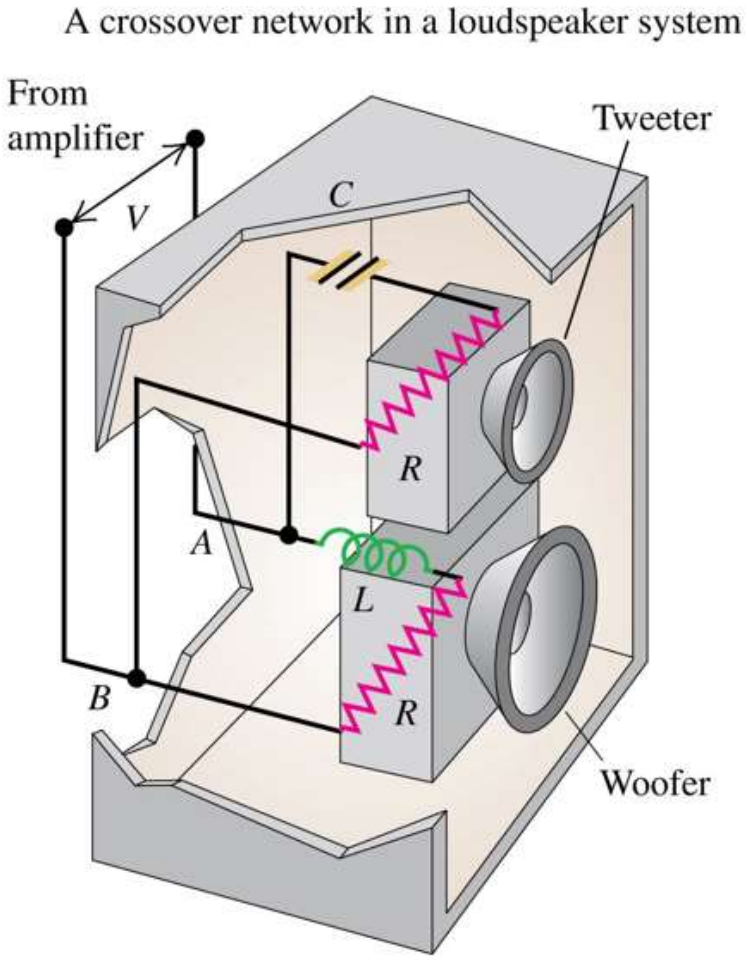


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A Useful Application: The Loudspeaker

- In order to route signals of different frequency to the appropriate speaker shown, the woofer and tweeter are connected in parallel across the amplifier output.
- The capacitor in the tweeter branch blocks the low-frequency components of sound but passes the higher frequencies.
- The inductor in the woofer branch blocks the high-frequency components of sound but passes the lower frequencies.

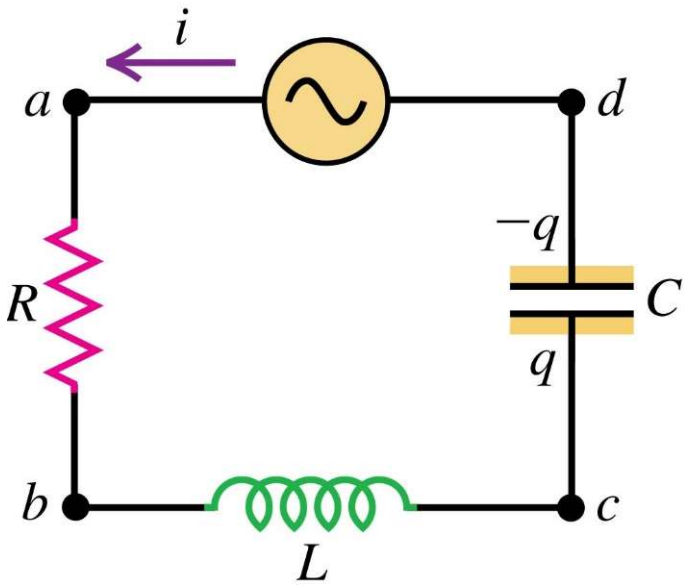


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The L-R-C Series Circuit (1 of 3)

- When a resistor, inductor, and capacitor are connected in series with an ac source, the voltage and current amplitudes are related by:



Amplitude of voltage across an ac circuit $V = IZ$ (I: Current amplitude; Z: Impedance of circuit)

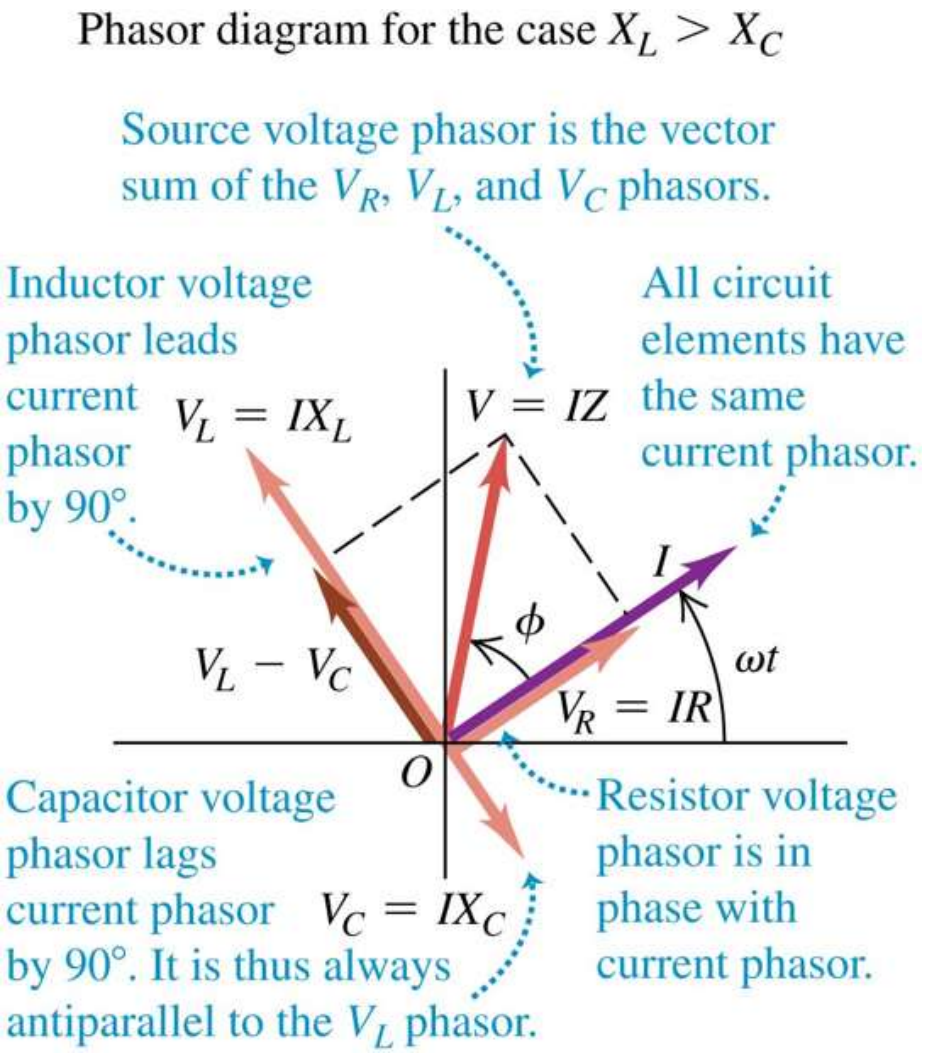
- The impedance of the circuit is:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

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The L-R-C Series Circuit (2 of 3)



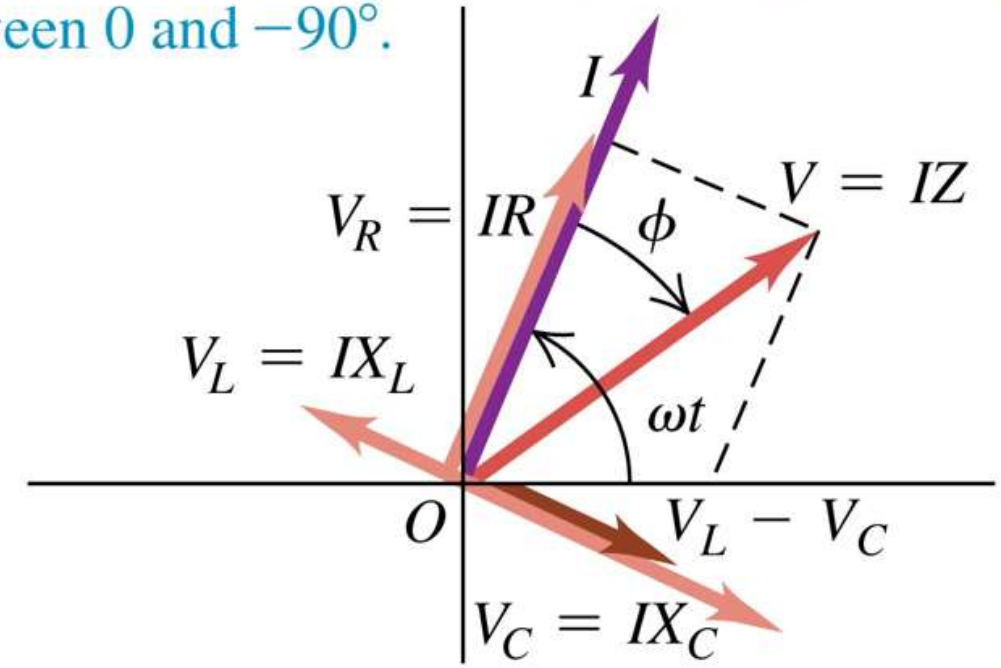
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The L-R-C Series Circuit (3 of 3)

Phasor diagram for the case $X_L < X_C$

If $X_L < X_C$, the source voltage phasor lags the current phasor, $X < 0$, and ϕ is a negative angle between 0 and $-90°$.



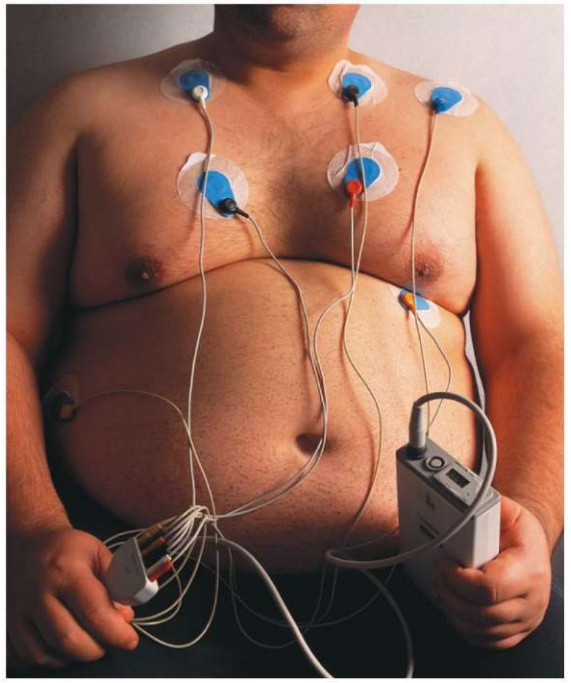
- Video Tutor Solution: Example 31.4

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Measuring Body Fat by Bioelectric Impedance Analysis

- The electrodes attached to this overweight patient's chest are applying a small ac voltage of frequency 50 kHz.
- The attached instrumentation measures the amplitude and phase angle of the resulting current through the patient's body.



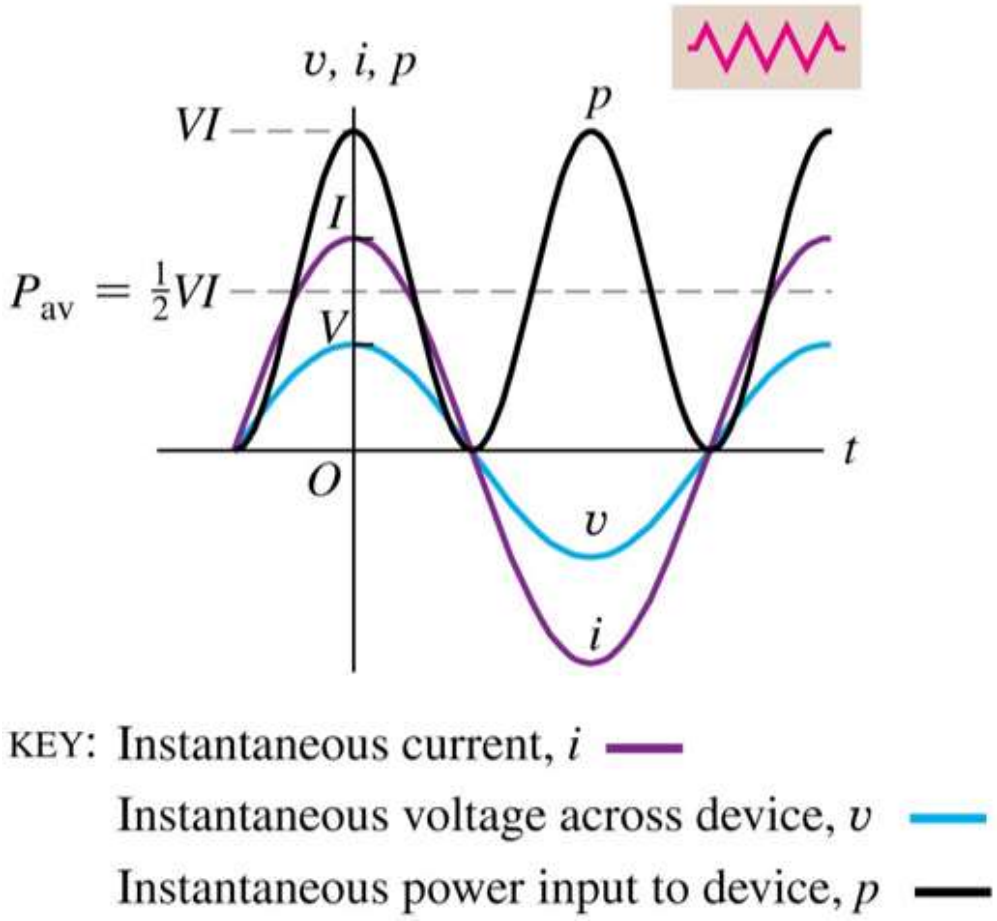
- These depend on the relative amounts of water and fat along the path followed by the current, and so provide a sensitive measure of body composition.

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Power in a Resistor

- If the circuit element is a pure resistor, the voltage and current are **in phase**.
- The instantaneous power $p = vi$ is always positive.

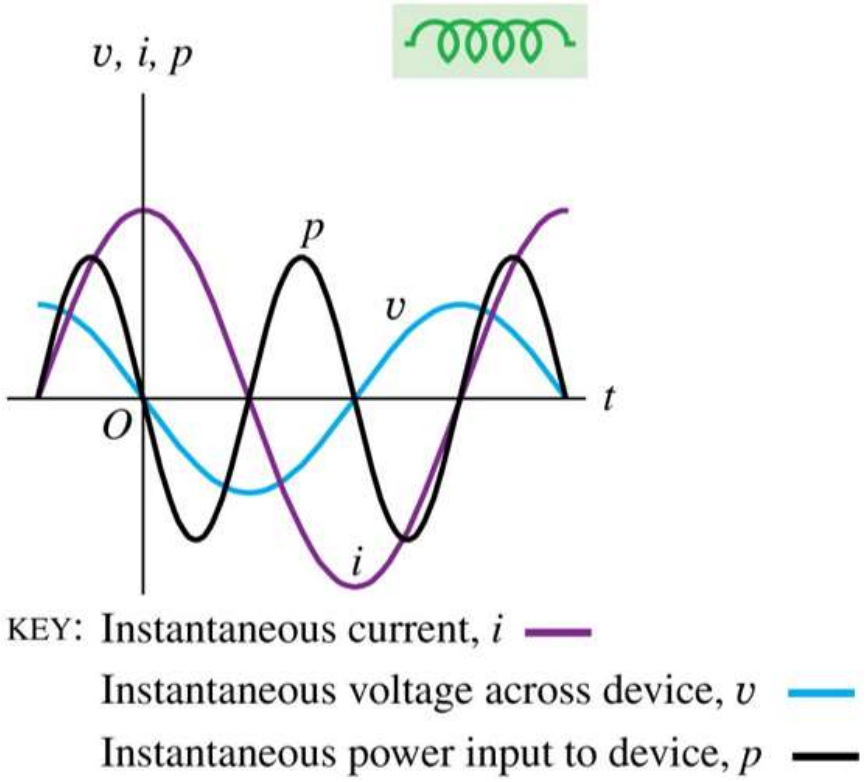


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Power in an Inductor

- If the circuit element is a pure inductor, the voltage **leads** the current by 90 degrees.
- The power is negative when *v* and *i* have opposite signs, and positive when they have the same signs.
- The average power is zero.

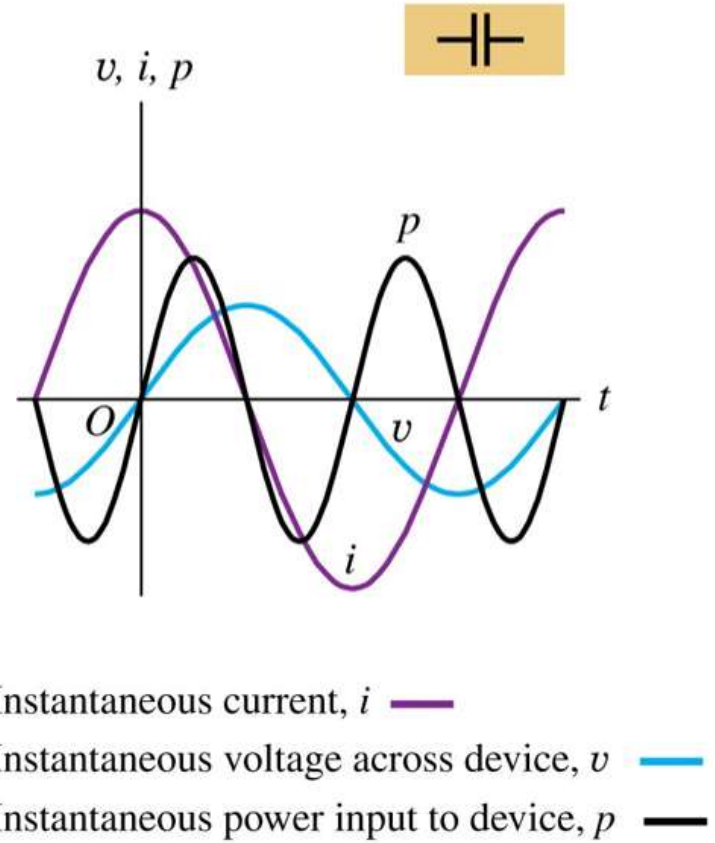


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Power in a Capacitor

- If the circuit element is a pure capacitor, the voltage **lags** the current by 90 degrees.
- The power is negative when *v* and *i* have opposite signs, and positive when they have the same signs.
- The average power is zero.

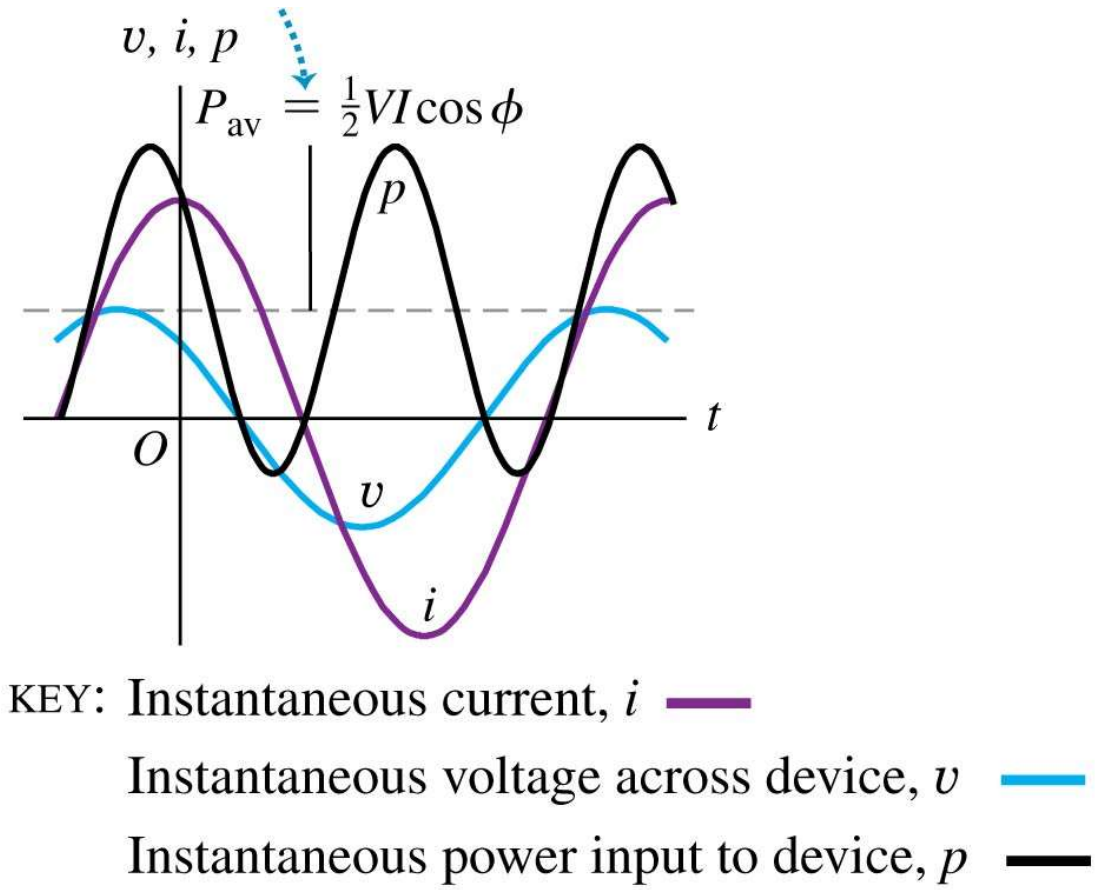


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Power in a General ac Circuit (1 of 2)

- For an arbitrary combination of resistors, inductors, and capacitors, the average power is positive.

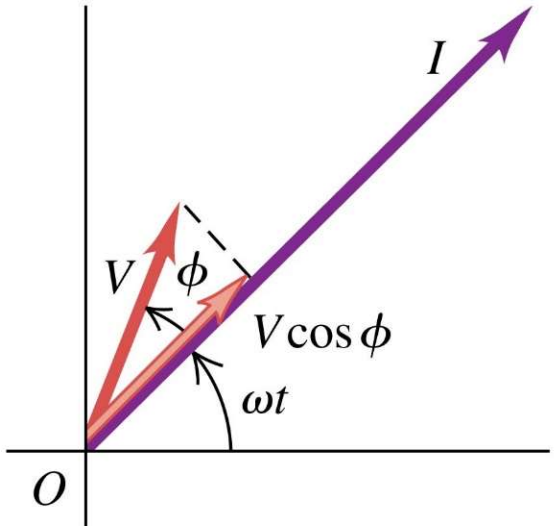


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Power in a General ac Circuit (2 of 2)

- In **any** ac circuit, with any combination of resistors, capacitors, and inductors, the voltage *v* across the entire circuit has some phase angle ϕ with respect to the current *i*.
- The factor cos ϕ is called the **power factor** of the circuit.



- For a pure resistor, the power factor is 1.

Phase angle of voltage with respect to current

Average power into a general ac circuit ⋯▸ $P_{av} = \frac{1}{2} VI \cos\phi = V_{rms} I_{rms} \cos\phi$

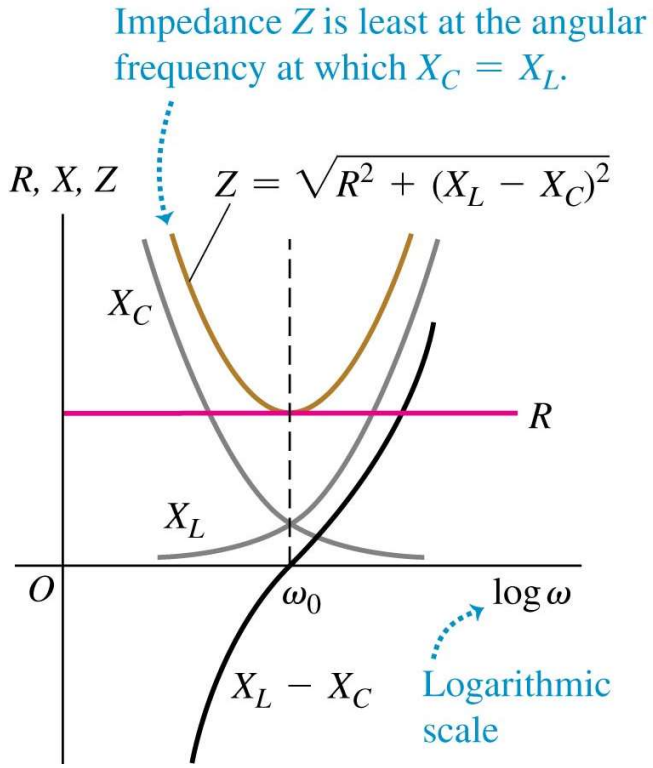
Voltage amplitude · Current amplitude · rms voltage · rms current

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Resonance in ac Circuits (1 of 3)

- Shown are graphs of R, X_L, X_C, and Z as functions of log ω.
- As the frequency increases, X_L increases and X_C decreases; hence there is always one frequency at which X_L and X_C are equal and $X_L - X_C$ is zero.
- At this frequency the impedance Z has its smallest value, equal simply to the resistance R.



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Resonance in ac Circuits (2 of 3)

- As we vary the angular frequency ω of the source, the maximum value of *I* occurs at the frequency at which the impedance *Z* is minimum.
- This peaking of the current amplitude at a certain frequency is called **resonance**.
- The angular frequency ω_0 at which the resonance Peak Occurs is called the resonance angular frequency.
- At $\omega = \omega_0$ the inductive reactance X_L and capacitive Reactance X_C are equal, so $\omega_0 L = \frac{1}{\omega_0 C}$ and:

Resonance angular frequency of an *L-R-C* series circuit

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Inductance, Capacitance

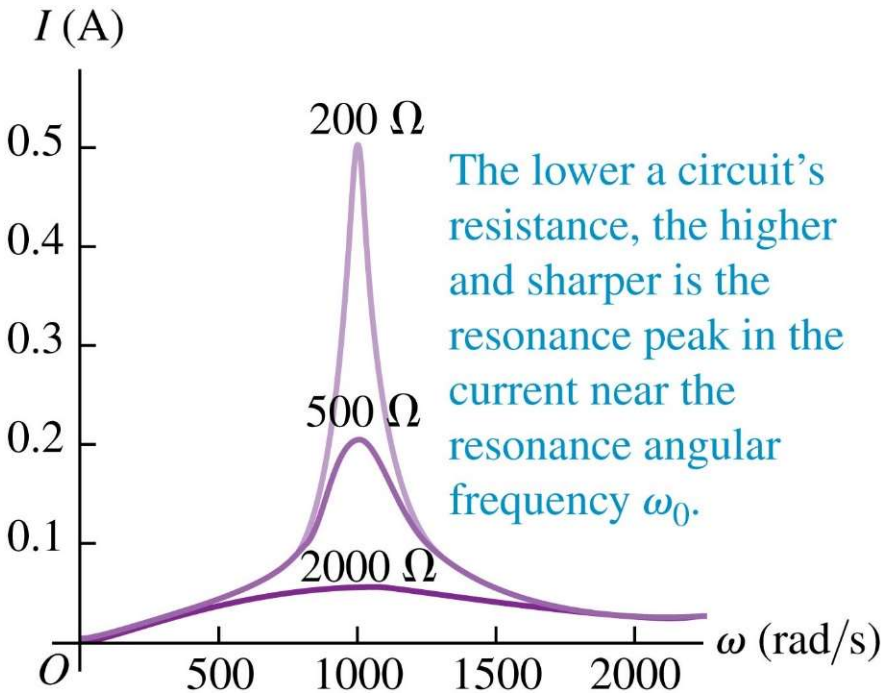
- Video Tutor Solution: Example 31.8

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Resonance in ac Circuits (3 of 3)

- Shown is a graph of current amplitude I as a function Of angular frequency ω for an L-R-C series circuit with V = 100 V, L = 2.0 H, C = 0.50 mF, and three different values of the resistance R.



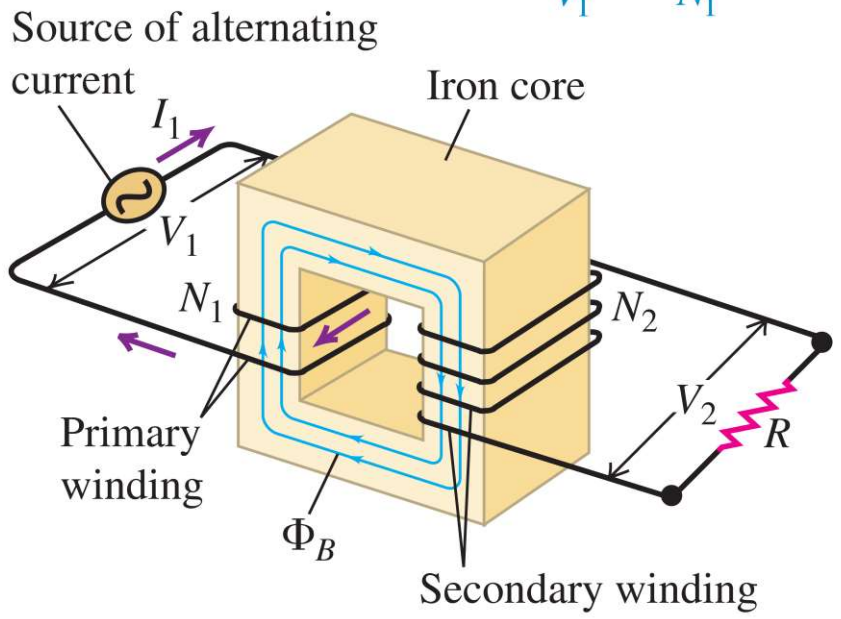
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Transformers (1 of 2)

- In a **transformer**, power is supplied to a primary coil, and then the secondary coil delivers power to a resistor.
- The purpose of a **step-up transformer**, such as the one shown, is to increase the delivered voltage relative to the supplied voltage.

The induced emf *per turn* is the same in both coils, so we adjust the ratio of terminal voltages by adjusting the ratio of turns: $\frac{V_2}{V_1} = \frac{N_2}{N_1}$



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Transformers (2 of 2)

- In an ideal transformer, the ratio of the voltages across the primary and secondary coils is equal to the ratio of the number of turns in the coils:

Terminal voltages in a transformer:

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

V_2: Secondary voltage amplitude or rms value

V_1: Primary voltage amplitude or rms value

N_2: Number of turns in secondary

N_1: Number of turns in primary

- If $N_2 > N_1$, then $V_2 > V_1$, and we have a **step-up** transformer.
- If $N_2 < N_1$, then $V_2 < V_1$, and we have a **step-down** transformer.

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End of Chapter

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